Grade 10 Term 1 Lesson 1 Practice Activity 2

Revenue, Costs, Profit, and Payoff Tables

Contents

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C3 Calculates profit as revenue minus costs across states of nature and organises the resulting payoffs in payoff tables.

1 Introduction

Problems:

1.1 Introductory General Problem (C1–C3)

A school enterprise must commit to either alternative A_1 or alternative A_2 before learning whether demand will be S_1 or S_2. Its accounting team used sales forecasts for each choice under each demand outcome to estimate total revenue, and used purchasing and operating budgets to estimate the corresponding total cost. All amounts and the resulting profit are measured in thousand USD.

If the enterprise selects A_1, forecast sales produce total revenue of 20 under demand state S_1 and 13 under S_2; fulfilling those sales is budgeted to cost 9 and 8, respectively. If it selects A_2, its different sales plan is expected to bring in 18 under S_1 and 15 under S_2, while the resources required by that plan cost 10 and 9, respectively. The enterprise must calculate the profit for every possible alternative–state pair before constructing its payoff table.

1.2 Advanced General Problem (C1–C3)

A community business must choose operating plan A_1 or A_2 before demand is revealed as S_1 or S_2. Its demand study indicates that plan A_1 will complete 2 units in state S_1 and 1 in state S_2, whereas plan A_2 will complete 3 and 2 units, respectively. All revenue rates, cost rates, fixed amounts, and profits are measured in thousand USD, with each variable rate applying to every unit completed.

The plans command different base selling rates because they offer different services: each unit under A_1 earns an alternative-based revenue rate of 10, while each unit under A_2 earns 11. Demand conditions add a market-based revenue rate of 5 per unit in S_1 or 3 per unit in S_2. The signed agreements also provide plan-based fixed revenue of 7 for A_1 and 9 for A_2, while community funding adds state-based fixed revenue of 6 in S_1 and 7 in S_2.

The business's operating budget assigns an alternative-based cost of 3 per unit to A_1 and 4 per unit to A_2, reflecting the resources used by each plan. Demand-related handling adds a further state-based cost of 1 per unit in S_1 and 2 per unit in S_2. Preparing the plans also creates fixed costs of 8 for A_1 and 10 for A_2, and state-specific administration creates fixed costs of 5 in S_1 and 3 in S_2. The business must combine all four revenue components and all four cost components for each pair, then subtract total cost from total revenue.

2 Direct Revenue and Cost Models

Problems:

2.1 Local Café Investment (C1–C3)

A small café must decide whether to launch a line of artisan desserts or keep its current menu before learning whether daily customer traffic will be high or low. The owner prepared direct total-revenue forecasts from expected café sales and direct total-cost budgets covering ingredients, staffing, preparation, and other resources. Every amount and the resulting daily profit are in thousand USD.

Launching the dessert line could generate revenue of 58 when traffic is high, when more customers buy the new products, and 27 when traffic is low. The matching operating budgets are 16 and 19; the low-traffic cost remains comparatively high because ingredients and preparation must be arranged before actual traffic is known. Keeping the current menu instead is forecast to produce revenue of 49 under high traffic and 41 under low traffic, with established-menu operating costs of 21 in either state. The owner must calculate profit for all four outcomes.

2.2 Green Energy Production Mix (C1–C3)

A renewable-energy company must build wind turbines, solar farms, or a balanced hybrid system before the year's weather is known. Annual electricity sales and the operating requirements of each generation system have been forecast directly for windy, sunny, and cloudy conditions. All revenue, cost, and profit figures are in thousand USD.

Wind turbines are expected to earn revenue of 132 in windy weather, 78 in sunny weather, and 62 in cloudy weather because their electricity output changes with conditions; the corresponding operating and maintenance costs are 42, 33, and 32. Solar farms instead are forecast to earn 77, 143, and 63 in the same respective states, with costs of 42, 48, and 38, reflecting the resources needed at each output level.

The hybrid system spreads production across both sources, producing forecast revenue of 112 when it is windy, 107 when it is sunny, and 96 when it is cloudy. Its corresponding operating costs are 42, 42, and 36. Management must subtract the appropriate direct cost from revenue for each strategy–weather combination.

3 One-Component Models

Problems:

3.1 Custom Printing Product Line (C1–C3)

A custom-printing cooperative must choose to produce premium notebooks or canvas bags before it knows whether the market will bring strong demand with stable material prices or weak demand with higher material prices. Orders already under discussion indicate that premium notebooks will sell 6 production lots in the strong state and 4 in the weak state, while canvas bags will sell 5 and 3 lots, respectively.

Because the two products have different selling prices, each notebook lot earns an alternative-based revenue rate of 12 thousand USD and each canvas-bag lot earns 15 thousand USD. Suppliers quote by market condition rather than product: stable prices make the cost 5 thousand USD per lot in the strong state, while higher prices raise it to 8 thousand USD per lot in the weak state. These are the only revenue and cost components, so the cooperative must apply both rates to the appropriate forecast lot quantity.

3.2 Event Equipment Fleet (C1–C3)

An event-equipment provider must prepare either a compact fleet or an expanded fleet before rental demand is known. Booking enquiries indicate that the compact fleet will fulfil 6 rental contracts under high demand and 3 under low demand; the expanded fleet has capacity for 8 and 4 contracts in those respective states.

Market prices make every completed contract earn 9 thousand USD when demand is high and 6 thousand USD when it is low, regardless of the fleet selected. Preparing, transporting, and inspecting the equipment requires a fixed outlay of 22 thousand USD for the compact fleet or 34 thousand USD for the expanded fleet. No other revenue or cost component is modeled, so the provider must compare state-based contract revenue with the chosen fleet's fixed preparation cost.

3.3 Community Arts Sponsorship (C1–C3)

A community arts organisation must accept either a standard or premium sponsorship agreement before officials confirm whether its event faces routine permit requirements or enhanced safety requirements. The sponsor guarantees a fixed payment of 52 thousand USD under the standard agreement or 68 thousand USD under the premium agreement, so this revenue depends only on the chosen agreement.

The authorities' requirements determine the event's fixed regulatory and safety bill: routine permits cost 18 thousand USD, while enhanced safety measures cost 31 thousand USD, whichever sponsorship agreement was selected. Because neither payment changes with attendance or output, no quantity is required; the organisation simply subtracts the relevant state-based fixed cost from its agreement-based fixed revenue.

3.4 Distribution Contract Fulfilment (C1–C3)

A distribution company must choose manual or automated packing before a new contract's volume is confirmed as standard or high. Capacity forecasts show manual packing processing 4 hundreds of orders under the standard-volume contract and 7 under the high-volume contract, while automated packing processes 5 and 8 hundreds, respectively.

The customer pays a fixed amount for the realised contract size: 70 thousand USD for standard volume or 104 thousand USD for high volume, regardless of the fulfilment method. Labour makes manual processing cost 7 thousand USD per hundred orders, whereas automation lowers that method-based rate to 5 thousand USD per hundred orders. With no other components, the company must deduct the chosen method's quantity-based processing cost from the state's fixed contract payment.

4 Two-Component Models

Problems:

4.1 Farm Produce Subscription Plans (C1–C3)

A farm cooperative must offer either a standard or premium produce box before seasonal demand is known to be strong or weak. Its subscription forecast gives the standard box 7 hundreds of subscriptions in the strong state and 4 in the weak state, and the premium box 6 and 3 hundreds, respectively.

The standard box's contents generate a plan-based revenue rate of 8 thousand USD per hundred subscriptions, while the premium contents generate 11. Strong market conditions add 3 thousand USD per hundred subscriptions to either plan's revenue; weak conditions add 1. Packing inputs cost 4 thousand USD per hundred standard subscriptions or 6 per hundred premium subscriptions. Finally, seasonal distribution arrangements create a fixed cost of 9 thousand USD under strong demand and 13 under weak demand. The cooperative must combine the two revenue rates, then deduct both the plan-based variable cost and the state-based fixed distribution cost.

4.2 Mobile Catering Configuration (C1–C3)

A mobile-catering company must prepare a compact catering station or a full mobile kitchen before learning whether the event will be busy with stable ingredient prices or quiet with higher ingredient prices. Booking forecasts indicate 8 hundreds of service orders for the compact station in the busy state and 4 in the quiet state, compared with 10 and 5 hundreds for the full kitchen.

Menu sales earn a configuration-based 7 thousand USD per hundred orders from the compact station or 9 from the full kitchen. Booking agreements add fixed revenue of 18 thousand USD for the compact configuration and 25 for the full configuration. Ingredients cost 3 thousand USD per hundred orders when prices are stable and 5 when prices are higher. In addition, setting up the compact station costs a fixed 20 thousand USD, while preparing the full kitchen costs 32. The company must retain each component's alternative- or state-based role when calculating profit.

4.3 Online Training Programme (C1–C3)

An online-training provider must offer its foundation or advanced programme before enrolment is known to come from corporate clients or individuals. Past enquiries imply 5 enrolment groups for the foundation programme under corporate enrolment and 3 under individual enrolment; the advanced programme expects 4 and 2 groups, respectively.

Tuition generates 14 thousand USD per foundation group or 20 thousand USD per advanced group. The enrolment channel also provides a fixed payment of 12 thousand USD for corporate enrolment or 7 for individual enrolment. Delivering a group costs 6 thousand USD in the corporate state and 8 in the individual state, while state-specific platform arrangements add fixed costs of 10 and 14, respectively. The provider must apply both state-based costs to the forecast groups while keeping the fixed amounts independent of quantity.

4.4 Community Venue Membership (C1–C3)

A community organisation must select standard or expanded venue membership before booking demand is known to be high or low. Reservations are forecast at 6 bookings for standard membership under high demand and 3 under low demand, while expanded membership enables 8 and 4 bookings, respectively.

The market supports revenue of 10 thousand USD per booking in the high-demand state and 7 in the low-demand state. A sponsor also contributes a fixed 15 thousand USD when the standard plan is selected or 22 for the expanded plan. Annual membership itself costs a fixed 28 thousand USD for the standard plan and 40 for the expanded plan, while permits add a state-based fixed cost of 8 under high demand or 13 under low demand. The organisation must calculate revenue and subtract both fixed cost sources without multiplying them by bookings.

4.5 Urban Delivery Fleet (C1–C3)

An urban-delivery company must deploy conventional vans or electric cargo bikes before conditions resolve into high demand with light traffic or low demand with heavy traffic. Route forecasts assign the vans 9 delivery batches in the first state and 5 in the second, while cargo bikes can complete 12 and 6 batches, respectively.

Customer contracts pay 8 thousand USD per batch in the high-demand state and 6 in the low-demand state. Meeting state targets also brings a fixed performance payment of 14 thousand USD with high demand or 8 with low demand. Van operation costs 3 thousand USD per batch and cargo-bike operation 2; traffic then adds 1 thousand USD per batch in light traffic or 3 in heavy traffic. The company must combine the fleet-based and traffic-based rates for every delivered batch before deducting costs.

4.6 Data-Service Contract (C1–C3)

A data-services company must choose standard or automated processing before a contract is confirmed as standard-volume or high-volume. Work estimates assign standard processing 4 batches under standard volume and 7 under high volume, while automation handles 5 and 9 batches, respectively.

The method agreement pays fixed revenue of 55 thousand USD for standard processing or 72 for automation. The client adds a service-level payment of 16 thousand USD in the standard-volume state and 9 in the high-volume state. Processing costs 6 thousand USD per batch with the standard method or 4 with automation, and implementation adds a method-based fixed cost of 18 thousand USD or 31, respectively. The company must keep both revenue payments fixed while applying only the processing rate to batch quantity.

5 Three-Component Models

Problems:

5.1 Regional Parcel Delivery (C1–C3)

A regional logistics company must select its standard or express delivery plan before conditions become high demand with light traffic or moderate demand with heavy traffic. Forecasts give the standard plan 8 hundreds of delivery batches in the first state and 5 in the second, and the express plan 10 and 6 hundreds, respectively.

The standard service earns 9 thousand USD per hundred batches and express service earns 11; demand conditions add 3 per hundred batches under high demand or 2 under moderate demand. Service agreements also pay a fixed 16 thousand USD for the standard plan and 24 for express. Operating costs are 4 thousand USD per hundred batches for standard and 5 for express, while traffic adds 1 per hundred batches when light and 3 when heavy. Distribution arrangements finally cost a fixed 10 thousand USD in the high-demand, light-traffic state and 15 in the moderate-demand, heavy-traffic state.

5.2 Solar Installation Programme (C1–C3)

A renewable-energy company must offer a standard-panel or high-efficiency-panel programme before demand and permits resolve into strong commercial demand or limited residential demand. It forecasts 7 installation batches for standard panels in the strong state and 4 in the limited state, compared with 6 and 3 batches for high-efficiency panels.

Standard panels earn 12 thousand USD per batch and high-efficiency panels 18, while the market adds 4 per batch in the strong state or 2 in the limited state. An incentive provides fixed revenue of 20 thousand USD under strong commercial demand and 11 under limited residential demand. Installation inputs cost 6 thousand USD per standard-panel batch or

8 per high-efficiency batch. Programme setup costs a fixed 18 thousand USD for standard panels or 25 for high-efficiency panels, and permits add fixed costs of 9 in the strong state or 14 in the limited state.

5.3 Hybrid Conference Production (C1–C3)

An events company must choose a standard or immersive hybrid-conference package before attendance and sponsorship are known to be strong or limited. Production forecasts allocate 6 event segments to the standard package in the strong state and 4 in the limited state, while the immersive package uses 8 and 5 segments, respectively.

Each standard segment earns 13 thousand USD and each immersive segment 16. Booking contracts add fixed revenue of 14 thousand USD for the standard package or 22 for the immersive package, while sponsors add a further fixed 18 in the strong state or 9 in the limited state. Delivery costs 5 thousand USD per segment under strong conditions and 7 under limited conditions. Package setup adds a fixed cost of 24 for standard or 37 for immersive, and licensing adds a state-based fixed cost of 8 or 13, respectively.

5.4 Cold-Storage Export Contract (C1–C3)

A food-export company must choose a modular or large cold-storage facility before export demand is strong or limited. Capacity and demand forecasts give the modular facility 9 storage batches under strong demand and 5 under limited demand, and the large facility 12 and 7 batches, respectively.

Export conditions set revenue at 10 thousand USD per batch under strong demand or 7 under limited demand. The chosen capacity earns fixed revenue of 20 thousand USD for the modular facility or 30 for the large facility, and the export programme adds fixed revenue of 16 in the strong state or 8 in the limited state. Operating rates are 4 thousand USD per modular batch and 3 per large-facility batch; energy adds 2 per batch under strong demand or 4 under limited demand. Preparing the configuration also costs a fixed 28 thousand USD for modular storage or 42 for large storage.

6 Complete Four-Component Models

Problems:

6.1 Smart Manufacturing Contract (C1–C3)

A manufacturer must select conventional or automated assembly before conditions become strong demand with favourable inputs or limited demand with expensive inputs. Forecast production is 8 batches for conventional assembly in the favourable state and 5 in the expensive-input state, versus 11 and 6 batches for automation.

Conventional batches earn a system-based 12 thousand USD each and automated batches 15; market conditions add 4 per batch in the favourable state or 2 in the expensive-input state. The contract also pays fixed revenue of 18 thousand USD for conventional assembly or 28 for automation, while a market incentive adds fixed revenue of 14 or 7 in the respective states.

Operating rates are 5 thousand USD per conventional batch and 6 per automated batch. Materials add 2 per batch under favourable inputs or 4 under expensive inputs. Setup costs are fixed at 24 thousand USD for conventional assembly and 38 for automation, while compliance adds a fixed 9 in the favourable state or 15 in the expensive-input state.

6.2 Distribution Centre Automation (C1–C3)

A retailer must choose manual, semi-automated, or robotic distribution before order volume and energy prices resolve into high volume with favourable energy or moderate volume with expensive energy. Forecast quantities, in hundreds of order batches, are 8 and 5 for manual distribution, 11 and 7 for semi-automation, and 14 and 8 for robotics across those respective states.

The configurations earn 12, 15, and 18 thousand USD per hundred order batches, respectively, because their service capabilities differ. The market adds 4 per hundred batches under favourable energy or 2 under expensive energy. Service contracts provide fixed revenue of 16, 24, and 35 thousand USD for manual, semi-automated, and robotic distribution, respectively; delivery incentives add fixed revenue of 15 in the high-volume state or 8 in the moderate-volume state.

Operating rates are 5, 6, and 8 thousand USD per hundred batches for the three respective configurations. Energy adds 2 per hundred batches in the favourable state or 4 in the expensive state. Installation creates fixed costs of 22, 34, and 50 thousand USD for manual, semi-automated, and robotic distribution, respectively, while compliance adds fixed costs of 9 under high volume or 16 under moderate volume.

6.3 Electric Vehicle Charging Network (C1–C3)

An energy company must build a neighbourhood or rapid-charging network before utilisation and electricity prices become high and favourable, moderate and standard, or low and expensive. Forecast quantities, in hundreds of charging-session batches, are 10, 7, and 4 for the neighbourhood network across those states, and 14, 9, and 5 for the rapid network.

Neighbourhood service earns a network-based 11 thousand USD per hundred charging-session batches and rapid charging earns 16. Utilisation conditions add 4, 3, and 2 thousand USD per hundred batches in the high, moderate, and low states, respectively. Network service agreements provide fixed revenue of 18 thousand USD for the neighbourhood network or 30 for rapid charging, while utilisation incentives add fixed revenue of 16, 10, and 5 in the respective states.

Equipment costs 4 thousand USD per hundred batches for the neighbourhood network or 7 for rapid charging. Electricity adds 2, 3, and 5 thousand USD per hundred batches in the respective states. Infrastructure creates a fixed cost of 25 thousand USD for the neighbourhood design or 45 for rapid charging, while state-dependent maintenance adds fixed costs of 8, 12, and 18, respectively. The company must preserve all eight components when computing each network–state profit.